ALERT meeting

Kalman Filter updates

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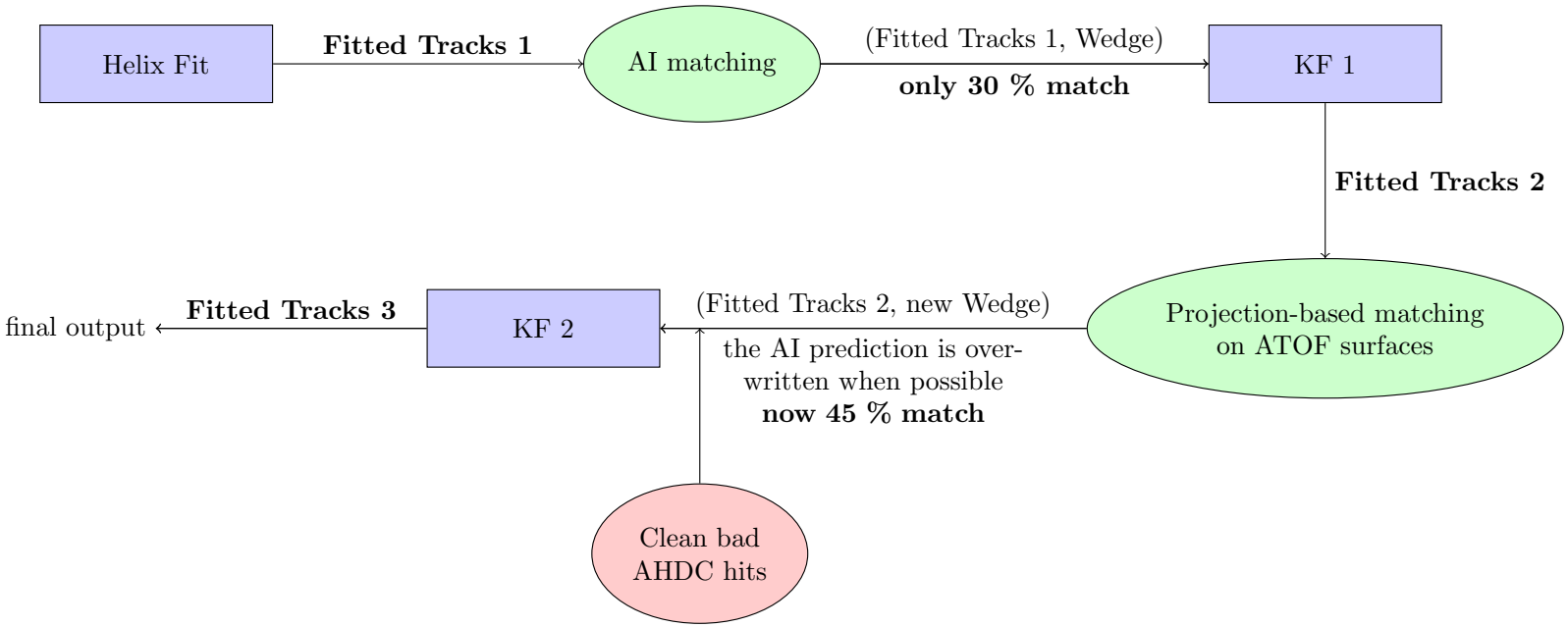
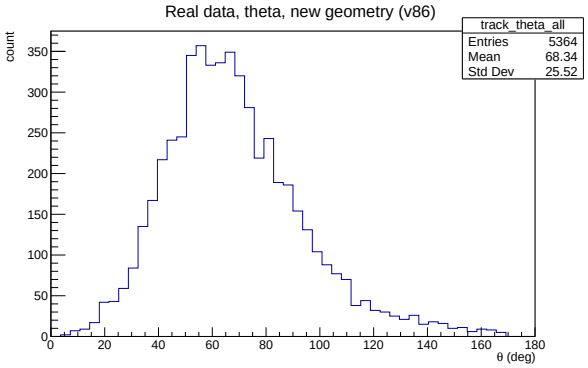
ATOF hit work completed 1/2

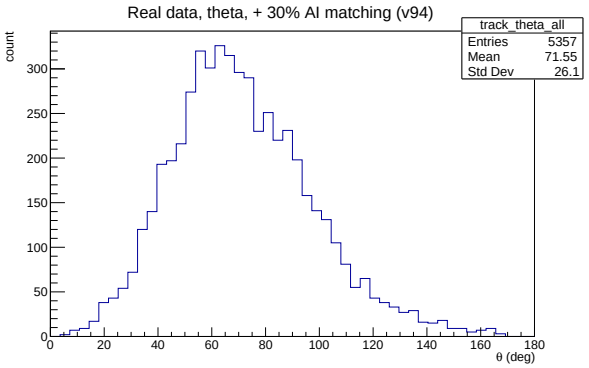
- The Implementation of the ATOF hits in the Kalman Filter is completed
- Current tracking strategy

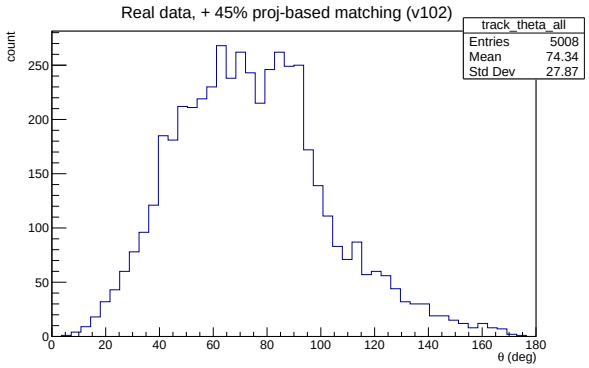


ATOF hit work completed 2/2

- Status of the theta reconstruction
- We have two peaks in the last version of the theta reconstruction
  - The first peak are tracks without ATOF matches
  - The second peak around 86° are tracks with ATOF matches (45 %)
- The final 45 % matches (over the nb. tracks) is low; it is essentially due the current misalignment between the AHDC and ATOF







Next steps

- Convenient use of the PID
  - For now, the energy loss and the curvature assume the particle to be a proton
  - We will use the PrePID work of Uditha (link)
- Material verification
  - Correct density for the target straw, target gas, ATOF scintillator
  - Correct use in the Kalman Filter propagator
  - **Vary the density of the target gas with the temperature and the pressure condition over the run period**
- Fine-tuning
  - Electron vertex resolution on p_e and θ_e
  - Measurement noise of the AHDC hits (ADC and time dependence)
  - ATOF hits resolution for clusters
  - Initial error covariance matrix
- Work on the AHDC alignment
  - With respect to CLAS
  - With respect to ATOF